

Date:27/10/2025

Title: Smart water storage tank

Aim: To design smart water storage system using CAE plus and 250 SCALA Programming that provides:

- Automatic Water control for tank filling and over flow prevention.
- High- and Low-Level Indication to monitor the tank water level status.

Objectives: To design smart water storage tank using CAE plus and 250 SCALA.

- At low level, when the water is at 10% of its volume the Pump gets ON.
- At high level, when water level is 90% the pump gets OFF.
- Alarm Indication for high and low level of water.

Theory

The **250 SCALA Programmable Logic Controller (PLC)** plays a central role in managing and automating the water level control system. Acting as the brain of the operation, it continuously monitors input signals from **water level sensors** strategically installed at key points within the storage tank — typically at the high and low water levels. These sensors are responsible for detecting the precise position of the water surface and transmitting that information to the PLC in real time.

Once the signals are received, the **control logic**, which is carefully programmed and tested using **CAEplus software**, determines the appropriate action to take. When the water level drops below the preset **low-level sensor**, the PLC immediately sends a command to **activate the pump**, allowing water to flow into the tank. As the water level gradually rises and reaches the **high-level sensor**, the PLC responds by **deactivating the pump**, effectively stopping the inflow to prevent overflow. This automatic control cycle repeats as needed, maintaining a stable and reliable water level within the system.

By incorporating **CAEplus** in the logic design process, engineers can **simulate, debug, and verify** the control program before deploying it to the 250 SCALA controller. This not only ensures a **smooth and error-free operation** but also enhances the **safety, efficiency, and reliability** of the system. The automation minimizes the need for manual supervision, reduces

energy waste, and prevents issues such as **pump dry running or tank overflow**, making it an essential solution for modern industrial and domestic water management applications.

Apparatus required

Sl.no	Components	Specification	Quantity
1	PLC	SIEMENS, SICAM A8000	1
2	Computer	Core i5, intel	1
3	PLC workbench Manual	SICAM A8000 workbench manual	1
4	Power supply (SMPS)	Omron S8VK-C24024, INPUT:100-240V, 50/60 Hz, 3.1A, Output: 24V DC, 10 A	1
5	Automation Test Bench	PLC (SICAM A8000) workbench	1
6	Software	TOOLBOX II, CAEX plus, SCADA server, Picture editor, SCADA client	5
7	Power supply	220V, 50Hz single phase	1
8	Input device	Analog input: R01 Digital inputs: S01, S02	4
9	Output device	Digital outputs H01, H02, H03, H04, H05,	5

Process Overview

The Smart Water Storage System is designed to automatically monitor and control the water level in a storage tank using a combination of 250 SCALA PLC, analog sensors, and a SCADA interface. The system ensures efficient water management by automatically operating the feeder pump based on real-time tank level signal, thereby preventing both overflow and dry-running conditions.

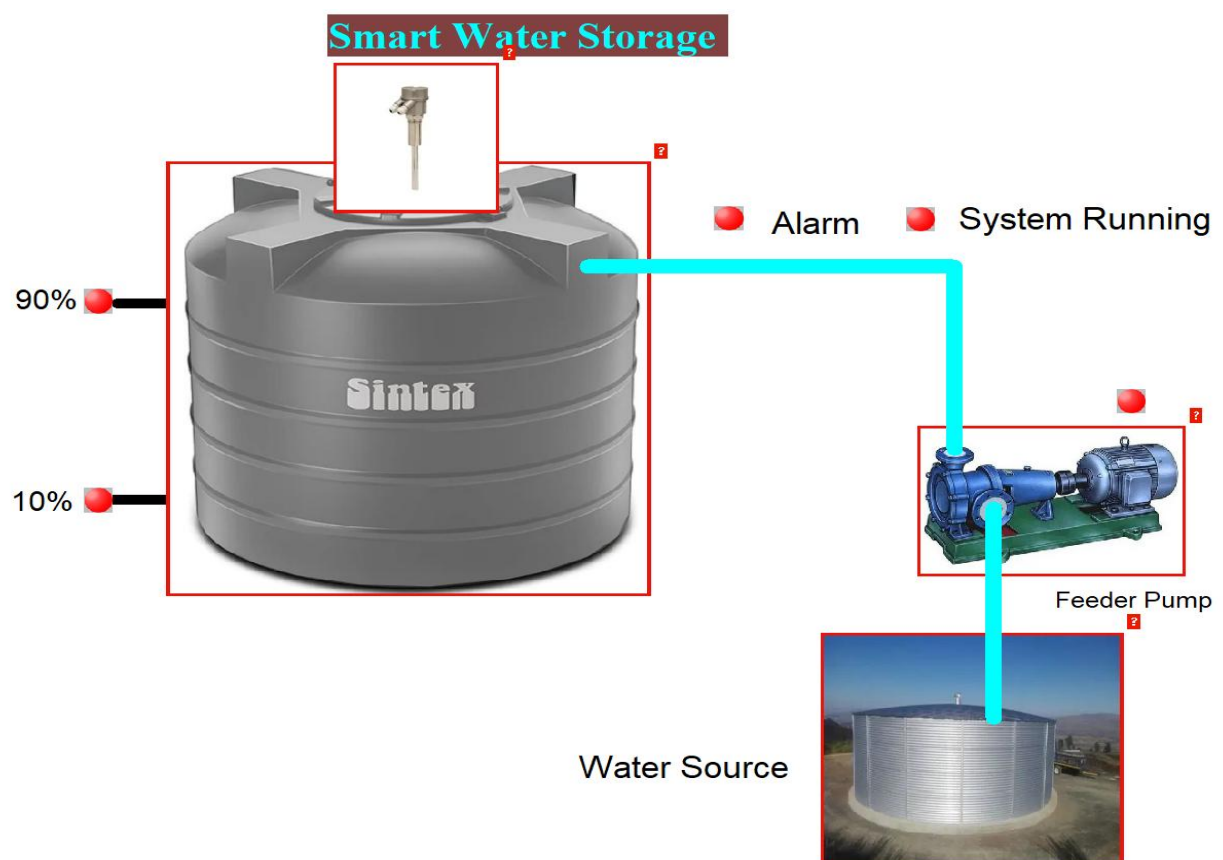
An analog water level sensor (AI. R01) continuously measures the tank's water height and transmits a 4–20 mA current signal to the PLC. The PLC then converts this signal into a percentage level, where 4 mA corresponds to an empty tank (0%) and 20 mA represents a full tank (100%). Based on this conversion, threshold limits are defined at 5.6 mA ($\approx 10\%$) for the **low-level** condition and 18.4 mA ($\approx 90\%$) for the **high-level** condition.

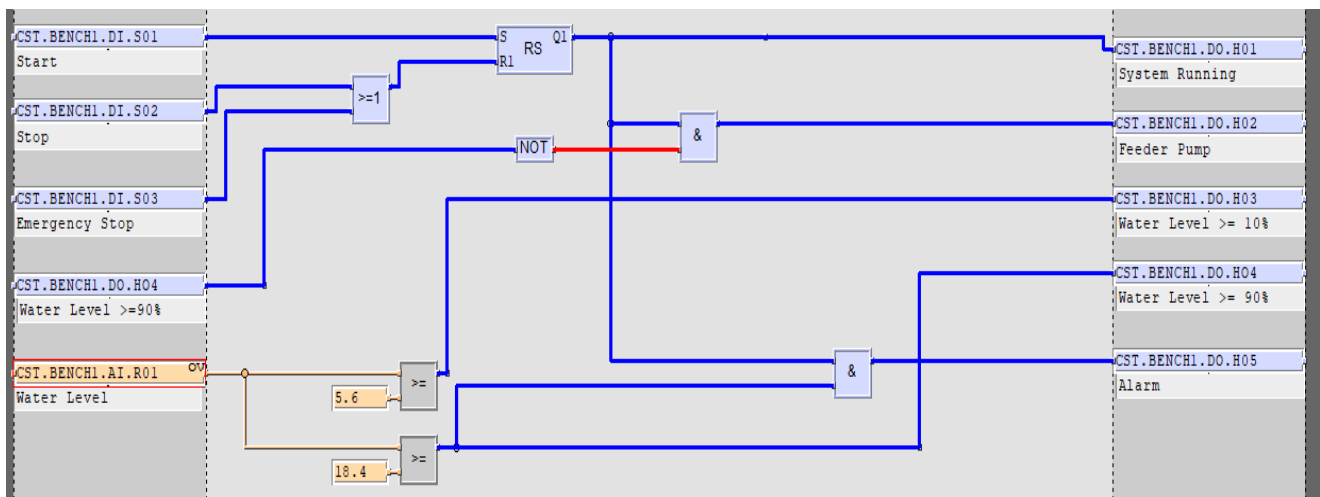
When the water level drops below 10%, the PLC logic automatically activates the feeder pump to fill the tank. Once the level reaches 90%, the pump is deactivated, preventing overflow. The system's operation is further controlled by **Start, Stop, and Emergency Stop** switches, which are implemented using **logic gates** and an RS flip-flop to maintain proper system latching and safety.

The entire process is visually monitored and controlled through a SCADA interface, designed using the Picture Editor and displayed on the SCADA client. The graphical display shows key elements such as the water tank, pump, and water source, Water Level Status, and Alarm. The alarm is triggered when the tank reaches the high-level threshold.

Overall, the system provides a fully automated and safe solution for water storage management. It eliminates manual supervision, ensures optimal pump operation, and enhances reliability through integrated PLC logic and SCADA visualization.

Circuit diagram





Procedure

Design the process in the picture editor

2. And then linked PLC and Picture editor

3. Create new instance

4. Calibrated the data for R01 as

- X 0% → 4
- X 100% → 20
- Y 0% → 0
- Y 100% → 100

5. Program the Logic and connect OPM II and workbench.

6. Generate the code

7. Load the generated code

8. Do online test

9. After the practical shift OPM to offline mode and hang up the connection between workbench and OPM

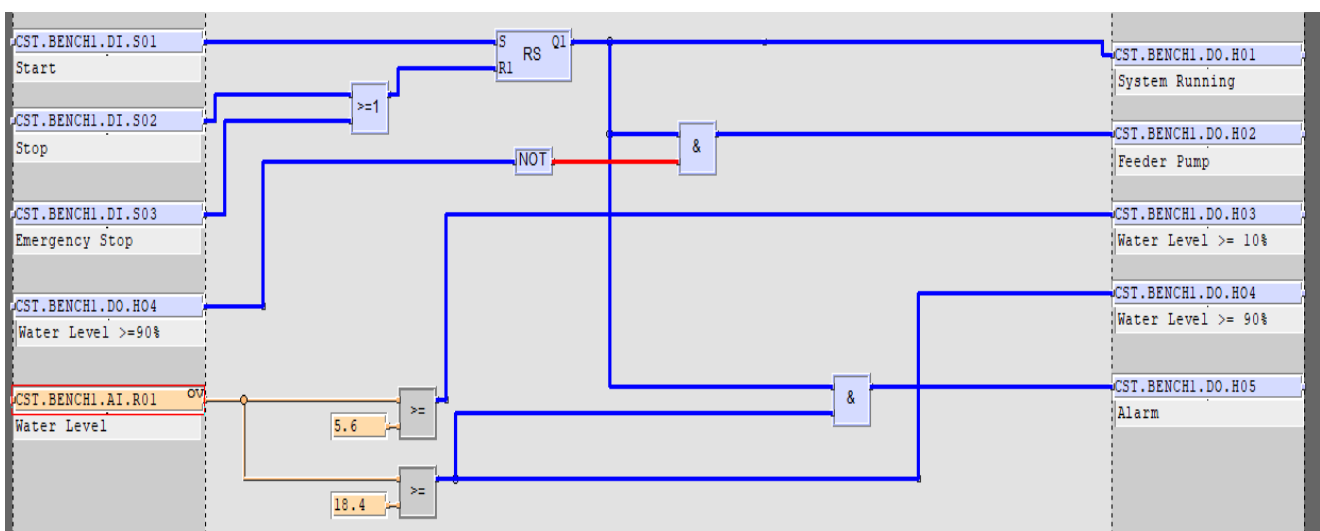
Precaution:

- Give power supply only when the circuit verification is done.
- Whenever we make slight change in the logic, always generate the code and load the parameters.
- Before doing offline test, make sure to run in online test mode.
- When importing pictures from other sources, the pictures must be in .jpeg format.
- Whenever we make any changes to the drawing, always save and update in the workspace.

Observation

Description of the Overall System Logic

The **Function Block Diagram (FBD)** shown below represents the **overall control logic** of the automated water storage system. It is implemented using the **250 SCALA PLC** and programmed through **CAEplus** software. The logic ensures automatic operation of the water feeder pump, system status indication, and alarm activation based on real-time water level readings.



Logic Components Used

RS Flip-Flop: Maintains the system ON/OFF state based on Start and Stop signals.

AND Gate: Ensures multiple conditions are true before activating an output (e.g., pump ON only when level <90% and system running).

NOT Gate: Inverts logic signals, used here to disable outputs when emergency or stop is triggered.

Comparator Blocks ($\geq$): Compare real sensor values with preset thresholds to determine level conditions.

The **analog water level sensor (AI. R01)** outputs a standard current signal ranging from **4 mA** to **20 mA**, which represents the **entire measurable height of the water tank** — from **0% (empty)** to **100% (full)**. This signal is compared against two threshold values:

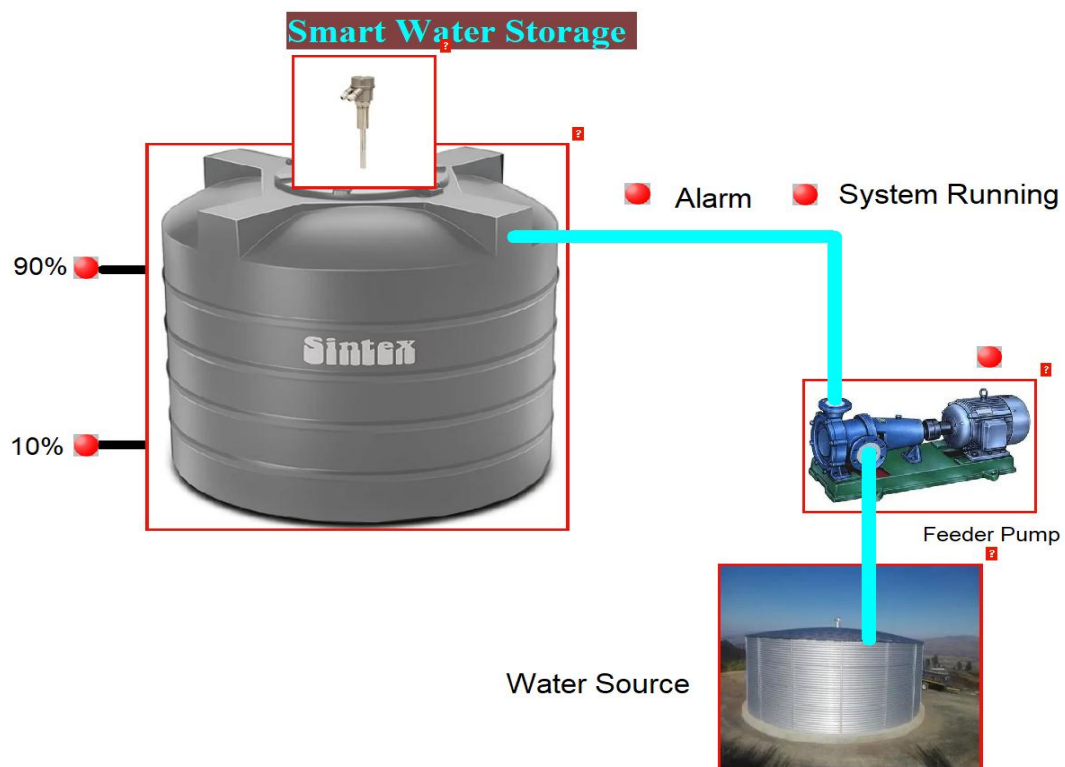
- 5.6 ($\approx 10\%$) – representing the low-level threshold.
- 18.4 ($\approx 90\%$) – representing the high-level threshold.

$$\text{Water level} = \frac{I_{\text{measured}} - 4}{20 - 4} \times 100$$

$$\text{Water level} = \frac{5.6 - 4}{20 - 4} \times 100 = 10\%$$

$$\text{Water level} = \frac{18.4 - 4}{20 - 4} \times 100 = 90\%$$

SCADA Design



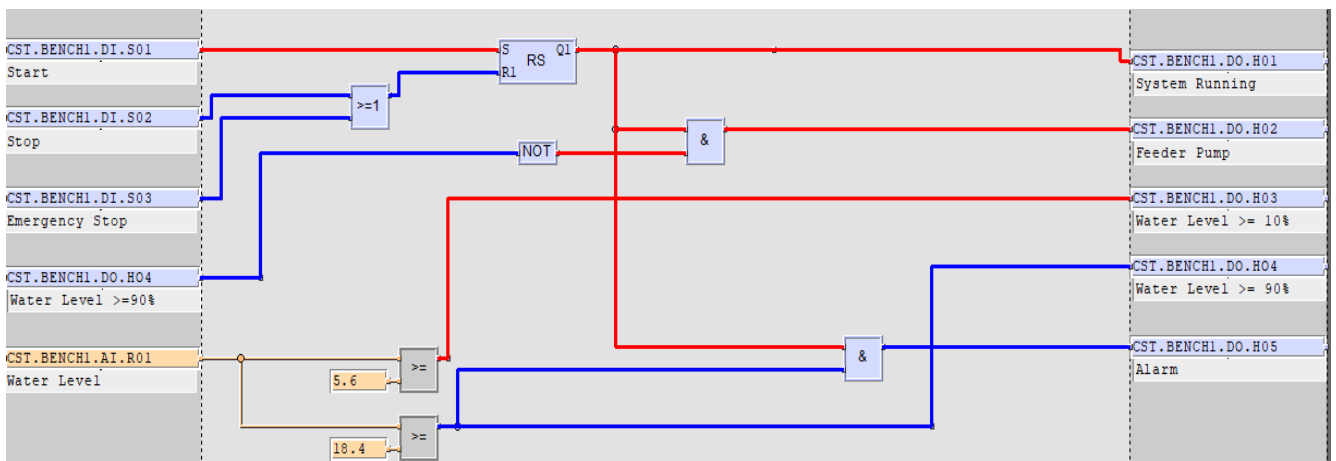
The above image illustrates the **SCADA (Supervisory Control and Data Acquisition)** interface developed for the **Smart Water Storage System**. The graphical display was created using the **Picture Editor** tool and deployed on the **SCADA Client workstation** for real-time

system monitoring and control. The design provides a clear visual representation of the entire water management process, showing the **water tank**, **feeder pump**, and **water source**, along with key status indicators.

Results

The system is **started** using the **Start button (DI. S01)**. When this input is high, it sets the **RS flip-flop**, turning ON the System Running indicator (**DO.H01**). The RS (Reset-Set) flip-flop holds the system in the ON state even if the Start button is released.

The analog water level sensor (AI. R01) continuously monitors the tank level and outputs a real-time signal. At first when the water level is at 10% the **feeder pump D0.H02** gets on to fill the storage tank until it reaches 90%.



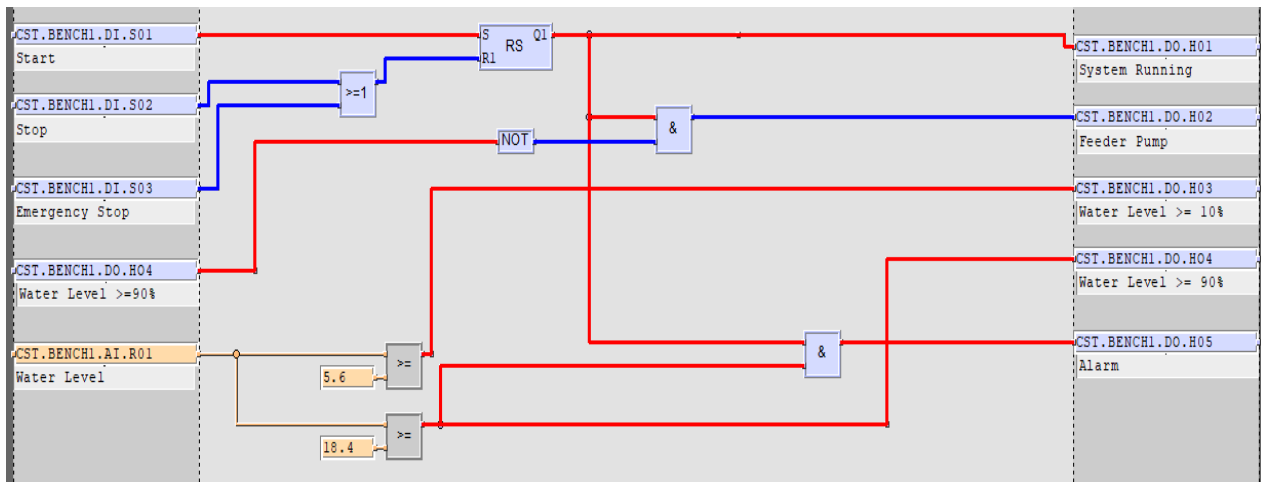
Feeder Pump Control

The **Feeder Pump (DO.H02)** is controlled through an **AND gate** that ensures it only operates under safe and required conditions.

The pump turns **ON** when, the system is running and the water level is **below 90%**, and the water level is **at or below 10%** (low threshold). This ensures automatic refilling when the tank is nearly empty while preventing overflow when it reaches 90%.

Level Indicators

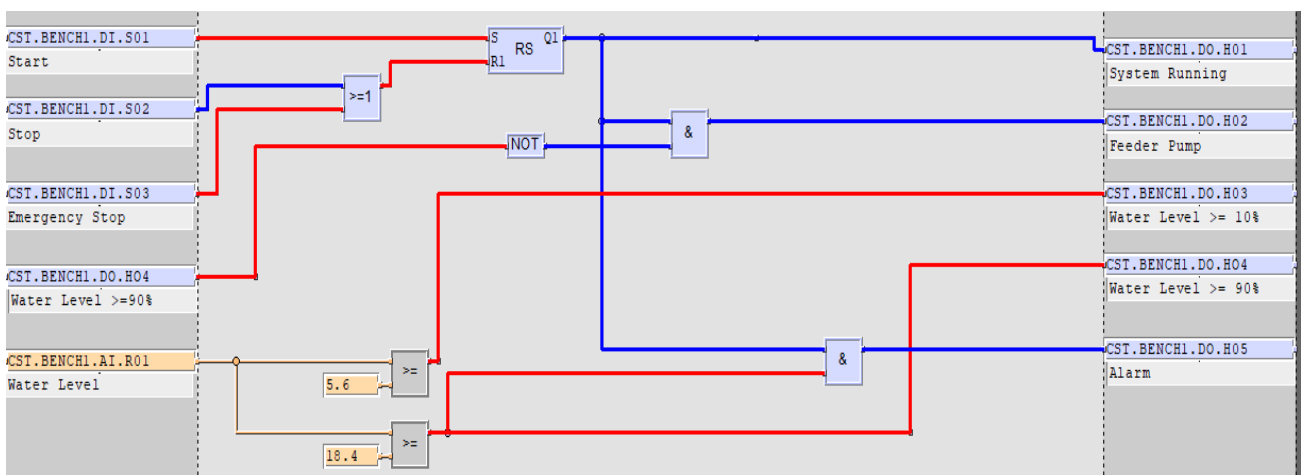
DO.H03 is activated when the water level reaches **10%**, indicating a low-water condition, hence the feeder pump gets On and the storage tank is filled until it reaches 90% of its total volume.



DO.H04 is triggered when the water level reaches **90%**, showing that the tank is nearly full. The **Alarm output (DO.H05)** is connected via an AND gate, which becomes active when the water reaches or exceeds the high-level mark. This serves as a visual or audible warning to indicate that the tank is full. At this state the feeder pump turns off.

This control logic provides a **fully automated water management system** that minimizes manual intervention, prevents overflow, and protects the pump from dry-running conditions. The use of **logic gates and flip-flops** ensures reliability and safety, while the analog level sensing enables accurate and continuous monitoring of tank status.

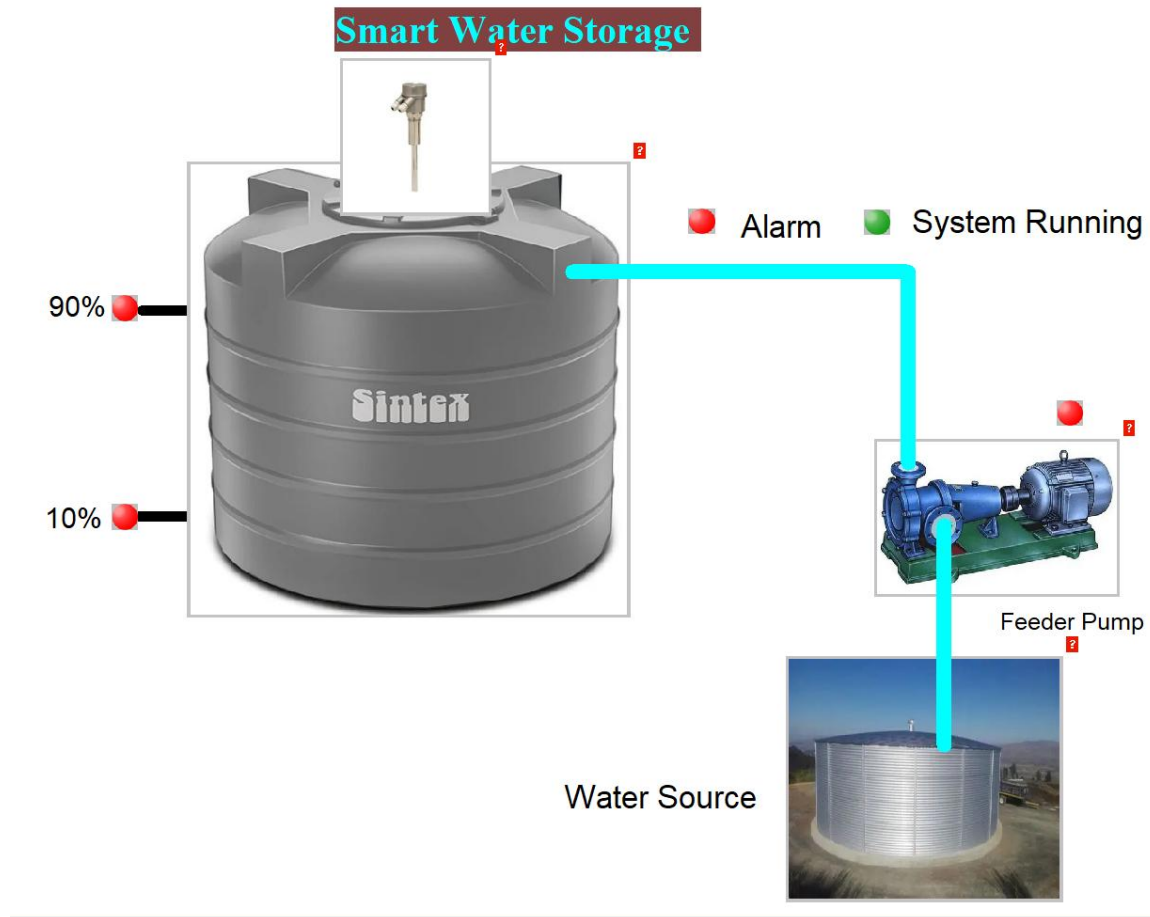
Stop Condition



The **Stop button (DI. S02)** and the **Emergency Stop (DI. S03)** act as reset inputs. When either is pressed, the RS flip-flop resets, immediately **stopping all operations** and turning off outputs. This control logic provides a **fully automated water management system** that

minimizes manual intervention, prevents overflow, and protects the pump from dry-running conditions. The use of **logic gates and flip-flops** ensures reliability and safety, while the analog level sensing enables accurate and continuous monitoring of tank status.

SCADA Interface



When the **Start button** is pressed, the system becomes active, automatic water control process begins. On the **SCADA interface**, this is visually represented by the **System Running indicator turning green**, while all other indicators remain red, signifying that they are currently off. This gives a clear visual confirmation that the control logic has been successfully initiated and that the system is now in standby, ready to respond to water level changes.

As the system runs, the **water level sensor** mounted at the tank opening continuously measures the current water level. This sensor sends a proportional **4–20 mA signal** to the PLC, allowing it to determine whether the tank is full, empty, or in between. Based on this signal, the **feeder pump** operates automatically according to the programmed conditions.



When the water level falls below the low threshold (around 10%), the pump starts to fill the tank. Once the level reaches the upper limit (around 90%), the pump stops, preventing overflow.

The **System Running** and **Alarm** indicators provide visual feedback of the system's operational status the System Running lamp turns ON when the process is active, while the Alarm lamp is triggered when the tank reaches its high-level limit.

This SCADA design ensures a user-friendly interface for operators to **monitor, control, and respond** to the system's behavior in real time. It enhances operational reliability and safety by allowing immediate visualization of process conditions and enabling rapid intervention in case of abnormal situations such as overfilling or pump failure.

Learning Outcomes

Through this practical, I gained a deeper understanding of how Programmable Logic Controllers (PLCs) and SCADA systems work together to achieve automation in industrial

processes. I learned how to design and implement control logic that responds to real-time sensor data, making the system self-operating and reliable. This experience strengthened my ability to interpret function block diagrams (FBDs) and understand how logic gates such as AND, OR, and NOT, along with RS flip-flops, contribute to process control.

Additionally, I learned how to scale analog signals (4–20 mA) into meaningful engineering values like water level percentages, and how to set threshold limits for automatic control actions. This helped me understand the relationship between sensor calibration and control response, ensuring that the system performs accurately under different operating conditions.

Lastly, this exercise enhanced my practical knowledge of industrial communication and visualization. I learned to create a SCADA interface that displays real-time process data and control status, improving both safety and usability. The integration of hardware logic and software visualization gave me a complete view of how automated systems operate in real industrial environments.

Discussion

This practical involved designing and simulating an automated water tank control system, providing a comprehensive, software-based exploration of the automation design lifecycle. The entire process was conducted through simulation, allowing us to focus on the principles of control logic and system visualization without physical hardware.

The project began within the CAEplus software environment, where the core control logic was meticulously designed and programmed. This initial phase was foundational, requiring a deep dive into the system's functional requirements to translate the operational aim—maintaining water levels between 10% and 90%—into a reliable and fail-safe program for the simulated PLC. Each software component, from basic logic gates (AND, OR) to more complex function blocks like the RS flip-flop and comparison blocks, was carefully selected and configured. The RS flip-flop was particularly crucial, as it demonstrated the concept of a "latching" mechanism, ensuring the simulated pump remained energized once started by the low-level signal until the high-level command definitively broke the circuit. A key learning outcome was orchestrating the interaction between the simulated digital command inputs (Start, Stop, and Emergency Stop) and the continuous analog signal from the virtual water level sensor. This required correctly scaling the analog input and ensuring the comparison blocks triggered actions at the precise thresholds, effectively modeling how a real PLC processes different signal types.

The final phase of the practical involved creating the operator interface using the 250 SCALA SCADA system's Picture Editor. Here, we built a dynamic graphical representation to visualize the simulation. This step focused on mapping the graphical objects—such as the animated tank level, the colour-changing pump symbol, and alarm indicators—directly to the specific tag addresses within the simulated PLC logic. Rigorous testing was then conducted to ensure robust data communication between the SCADA software and the simulation environment. Observing the graphical interface update in real-time, accurately reflecting the states and values generated by our control logic, was the culminating step that validated the entire simulated system's integration and functionality.

In summary, this practical was a valuable exercise in the software side of automation engineering. It allowed us to methodically apply control theory, from PLC programming in CAEplus to HMI development in 250 SCALA, within a simulated environment. The process solidified our understanding of how logic and visualization are integrated to create a complete and functional automation system model.

Application

The **Smart Water Storage System** has a wide range of practical applications across both **domestic** and **industrial** sectors. In residential settings, it can be used for **automatic water tank management**, ensuring that household water storage tanks are always maintained at optimal levels without the need for manual monitoring. This helps prevent common issues such as **overflow**, **dry running of pumps**, and **unnecessary power consumption**, thereby promoting energy efficiency and water conservation.

In industrial and commercial environments, the system can be integrated into **process plants**, **water treatment facilities**, and **irrigation systems**, where maintaining consistent water levels is critical for smooth operation. The use of PLC-based control provides **high reliability**, **accuracy**, and **flexibility**, allowing operators to adjust threshold levels and system parameters as per process requirements. With the addition of **SCADA monitoring**, operators can easily visualize system performance, receive alarms, and make informed decisions in real time.

Moreover, this system concept can be extended to larger **smart infrastructure projects**, such as **municipal water supply networks**, **hydropower plant reservoirs**, and **greenhouse irrigation systems**. In these applications, the combination of **sensor technology**, **automation**

logic, and remote monitoring can ensure sustainable and efficient water resource management. The modular design also allows the system to be scaled up or integrated with IoT platforms for advanced data analytics and predictive maintenance.

Conclusion

In this practical, we successfully implemented and observed the working of a **Smart Water Storage System** using a **250 SCALA PLC** and **SCADA interface**. The experiment helped us understand how automation can be applied to maintain a desired water level in a storage tank without manual supervision. By using the **4–20 mA analog input** from the water level sensor, the PLC was able to continuously monitor the tank's water level and control the **feeder pump** based on the programmed threshold values. When the level dropped below 10%, the pump automatically turned on, and when it reached 90%, it turned off — demonstrating reliable and accurate control logic.

We also learned how to develop the **logic in CAEplus software** using function blocks such as **AND, OR, NOT**, and **RS flip-flop** to achieve the desired control sequence. Additionally, the **SCADA design** provided a clear visualization of the process, showing the system's real-time status through indicators and alarms. This made it easy to observe how the system responded to different water level conditions and how the pump operated accordingly.

Overall, the practical gave us valuable hands-on experience with **PLC programming, hardware interfacing, and SCADA configuration**. It reinforced our understanding of industrial control systems and demonstrated the importance of automation in improving accuracy, efficiency, and safety. The exercise provided a strong foundation for more advanced applications of PLC and SCADA in future industrial or academic work.